Arjun Garg

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Technical Skills

Mechanical: SolidWorks, AutoCAD (orthographic drawings), Tinkercad, 3D printing, rapid prototyping, gearbox design
Electrical: circuit board prototyping, motor and circuit integration, root-cause failure analysis, Altium (learning)
Software: Python, C, C++, JavaScript, HTML/CSS, MATLAB (learning), Git/GitHub, VS Code, LaTeX
Certifications: National Lifeguard, Standard First Aid & CPR-C

Engineering Experience

Dynamics & Electrical Subteam Member 2026 – Present
Midnight Sun Solar Car Team & Baja Racing (Electrical), University of Waterloo *Waterloo, ON*

- Designing the solar car's parking brake in SolidWorks, developing it from hand sketches into a full 3D model

Team Engineer Summer 2025
SHAD Program – Memorial University *St. John's, NL*

- Calculated the motor RPM band for peak efficiency below 60 km/h on a three-wheeled urban vehicle
- Built a working mini-prototype by iterating on motors, custom circuit boards, CAD models, and 3D-printed parts, and learned KiCad to clean up the PCB layout; won Best Prototype
- Redesigned the gearbox housing after diagnosing vibration-induced flex that misaligned the gears
- Led design of an underwater ROV for a timed STEM competition; won with the fastest run

General Member 2024 – 2025
FIRST Robotics Team 1325, Inverse Paradox *Mississauga, ON*

- Diagnosed intermittent drive-shaft failures by isolating mechanical, electrical, and software causes
- Traced a recurring in-match shutdown to controller code commanding more output than the electrical system could deliver, ruling out an apparent mechanical fault; team won 4 awards and reached Provincial District Finals

Projects

Personal Portfolio Website | *HTML, CSS, JavaScript, Git* Fall 2026

- Developed and deployed arjungarg.me (projects, experience, interactive map), version-controlled on GitHub

Work Experience

Swim Instructor & Lifeguard 2025 – 2026
Life Time Fitness *Mississauga, ON*

- Taught twice-weekly swim lessons tailored to each child's level and served as a pool first responder

Leadership & Activities

Founder & President 2016 – Present
CandyForKids.org *Mississauga, ON*

- Scaled a youth-run charity from door-to-door drives to a GTA donation-box network and pickup service, distributing 2,000+ candy bags and saving ∼20,000 candies from waste
- Coordinated a 6-person executive team to reroute a 200+ bag distribution within 2 days after the partner food bank withdrew, setting criteria (proximity, legitimacy, capacity) to select a new recipient

General Executive 2024 – 2026
Peer Mentoring, John Fraser Secondary School *Mississauga, ON*

- Mentored 5 students through weekly check-ins and action plans; emceed a 100+ attendee Grade 9 Social

Competitor 2022 – 2026
DECA *Mississauga, ON*

- Qualified for the International Career Development Conference (2026); placed Top 20 Overall at Provincials

Education

University of Waterloo Sept. 2026 – Expected 2031
B.A.Sc. in Mechatronics Engineering, Stream 8 (Co-op) *Waterloo, ON*